ICT Course: Introduction to Cryptography

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Session 6: Asymmetric Cryptography - RSA Cryptosystem and Diffie-Hellman Key Exchange

1. RSA Cryptosystem
   - Generating RSA public and private keypair
   - Encryption-Decryption
   - Proof of the correctness
   - Repeated squaring
   - Speed up RSA
   - Cryptanalysis
2. Diffie-Hellman Key exchange
3. Application of Public Key cryptography

Overview of RSA

- be made practical by Rivest, Shamir and Aldeman
- the most widely used asymmetric cryptosystem
- Applications: Transport of (symmetric) keys and Digital Signature

RSA Key Generation

- Choose 2 large prime numbers p, q: $N = p * q$
- Choose e relative prime to $(p-1)(q-1)$
- Find d:

$$ed = 1 \; mod(p-1)(q-1) \iff e^{-1} \; mod(p-1)(q-1) = d$$

- RSA key pair consists of:
  - Public key: (N, e)
  - Private key: d

  where N: modulus, e: encryption exponent, d: decryption exponent
- N has 1024 bits or 2048 bits or larger

Encryption - Decryption

- Encryption: $C \equiv M^e (mod N)$
- Decryption: $M \equiv C^d (mod N)$

Proof of correctness

Euler's phi function reminds:

- $\phi(m)$ is the number of positive integers less than m that are relatively prime to m
- For any prime number p: $\phi(p) = (p-1)$
- If p, q are prime: $\phi(p * q) = (p-1)(q-1)$

Euler's Theorem

If x is relative prime to n then $x^{\phi(n)} \equiv 1 \; mod \; n$

Proof of correctness

- Assume that M is relatively prime to N, proof the correctness of RSA?
- If M is not prime to N, proof of the correctness of RSA?

Textbook RSA example

- $p = 11$, $q = 3$, choose $e = 3$
- Key pairs?
- Describe the encryption and decryption using RSA if Bob want to send a plaintext $M = 15$ to Alice?

- Example: $(N, e) = (33, 23)$, $M = 5$, Calculate C
- Problem?
- Repeated Squaring method

- Using same exponent e for all users and different p, q are chosen for each key pair
- Common used encryption exponent:
  - $e = 3$: requires $M > N^{1/3}$ to avoid **cube root attack**
  - $e = 2^{16} + 1$

Cryptanalysis

- Protocol attack
- Mathematical attack
- Side-channel attack

Diffie-Hellman Key exchange - Overview[1]

- Proposed in 1976 by Whitfield Diffie and Martin Hellman
- Widely used, e.g. in Secure Shell (SSH), Transport Layer security (TLS), Internet Protocol Security (IPSec)
- Used to establish a shared key, not usually for encryption

[1] Understanding Cryptography by Christof Paar and Jan Pelzl

Diffie-Hellman Key Exchange

Discrete Logarithm problem:

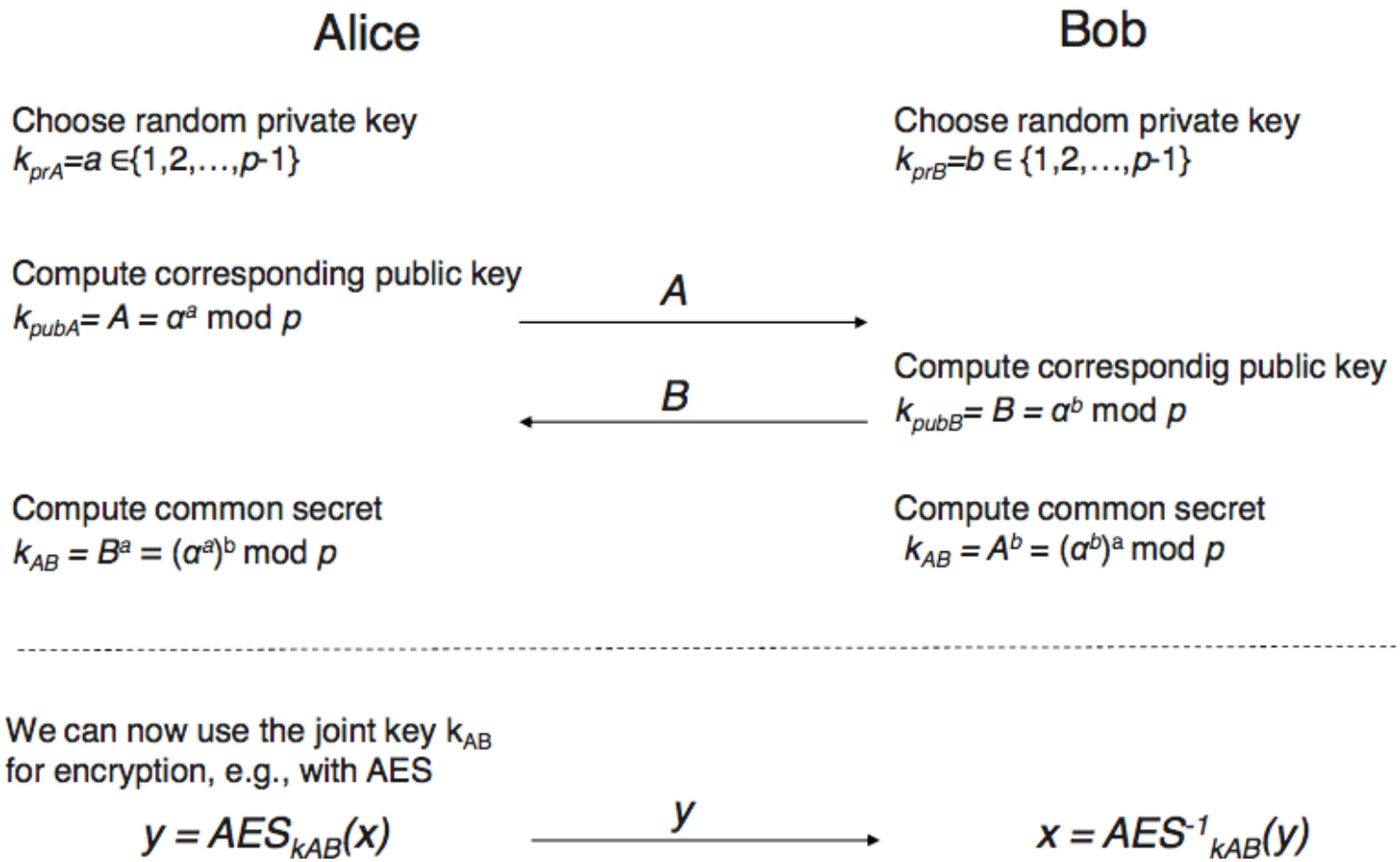
Given integers g, p, $g^k \ mod \ p$, find k

$\Longrightarrow$ Very difficult to solve

Diffie-Hellman Key Echange setup:

- Choose a large prime p and a (integer) generator g
- $\forall x \in \{1, 2, ..., p-1\}, \exists n : \ x \equiv g^n (mod \ p)$
- g, p are public

Diffie-Hellman Key Exchange(DHKE)



DHKE - Man-in-Middle Attack

- Confidentiality:
  - use key pairs to encrypt data
  - hybrid cryptosystem
- Integrity: Digital signature